

Vimal Harihar

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Education

Amrita Vishwa Vidyapeetham, Coimbatore
B.Tech in Computer Science and Engineering

Aug 2023 – May 2027
CGPA: 7.65 / 10.0

Coursework: Data Structures & Algorithms, Database Systems, Distributed Systems, Operating Systems, Software Engineering, Computer Networks

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL

CS Fundamentals: Data Structures, Algorithms, Object-Oriented Design, System Design, Problem Solving, Complexity Analysis

Full-Stack & UI/UX: React, Next.js, Node.js, FastAPI, REST APIs, WebSockets

Data & Cloud: SQL (MySQL, PostgreSQL), NoSQL (MongoDB, DynamoDB), Pandas, Data Pipelines, AWS, Azure, Docker, Git

Experience

Software Intern – Innoboorn

Apr 2026 – June 2026

- Developed an AI-driven **JIRA–GCP incident correlation system** using **RAG and Deep Agents** to identify relevant GCP audit logs for JIRA tickets and assist in production issue investigation.
- Built a **multi-stage retrieval pipeline** using metadata, keyword matching, BM25, and embeddings to rank relevant logs and provide contextual evidence for AI-based root-cause analysis.

Web Development Lead – Tensor AI Club

Mar 2024 – Jan 2026

- Led **20 developers** building the official platform for the Tensor AI Club, owning UI/UX, architecture, and delivery using React/Next.js with a focus on clean, responsive, high-quality interfaces.

Projects

Phanovex – Predictive Carry-Back Intelligence for Mining Trucks *Computer Vision, SAM3, Sensor Fusion, Python*
2026

- Caterpillar Tech Challenge 2026 Finalist:** Built a predictive intelligence system for CAT mining haul trucks that detects residual carry-back material in the truck bed, applying a **biomimetic design approach** to model how material adheres and detaches.
- Integrated the **SAM3 (Segment Anything) vision model** to segment and identify residual material from bed imagery, running inference to estimate buildup and flag carry-back risk before the next load.
- Engineered a **multi-sensor fusion layer** combining vision, load, and environmental sensor inputs, and built a **3D model and simulation** of the truck bed to analyze material behavior and validate the detection approach.

Lazarus – Cloud-Native Multi-Agent Platform on AWS *Python, AWS (Lambda, DynamoDB, S3), FastAPI* 2026

- Built a **cloud-native platform on AWS** using **Lambda** for serverless compute, **DynamoDB (NoSQL)** for state storage, and **S3** for object storage, orchestrating a multi-agent pipeline in which each agent handles one stage of the workload, with state persisted between stages.
- Designed **resilient REST APIs** with retry logic and failure recovery across Docker-containerized services.

Lekhaflow – Real-Time Collaborative Web Platform

React, TypeScript, Node.js, WebSockets

2025

- Built a **real-time web application** with a Node.js and TypeScript backend, using binary WebSocket serialization for low-latency state sync and a responsive React frontend with layered canvas rendering.
- Engineered **CRDT-based conflict-free state synchronization** across concurrent users, resolving simultaneous edits without central locking and keeping client replicas convergent under network delay and reconnection.

Open Source

REZO – Published a developer productivity extension on the VS Code Marketplace (live, publicly available).

Achievements

Finalist – Caterpillar Tech Challenge (2026) | **Winner** – Infineon PSOC6 Hackathon (2025) | **Finalist** – AWS AI for Bharat Hackathon (2026) | **Top 50/650** – NIIT Ideathon | **Top 50** – Smart India Hackathon

Certifications & Publications

Microsoft Learn: Azure Architecture & Cloud Infrastructure (AZ-305 & AZ-900 learning paths) | **LangChain Deep Agents, LangGraph, Introduction to LLMs** [Google]

AI's Influence on Gender Representation and Societal Norms in Emerging Technologies – *IEEE Published* | SecureAI-Cyber: An AI-Powered Cybersecurity Solution for Scalable Threat Management – *Published 2026*